SDV Resource Governor -- Technical Pitch Deck (8 Slides)

Slide 1: Executive Overview -- Deterministic Physics First

- **Mission**: Govern admission, placement, live arbitration, and graceful degradation across **8 coupled physical resource classes** (‘power‘, ‘cooling‘, ‘positions‘, ‘mast‘, ‘mass_stability‘, ‘data‘, ‘compute‘, ‘time‘) on a Software-Defined Vessel.
- **Core Principle**: Hard physical and life/vehicle safety limits are satisfied **first**; mission capability score, free margin, and displacement cost are optimized **second**. No priority weight can ever override a hard physical limit or drone ditching deadline.
- **Zero Opaque Runtime AI**: 100% deterministic, inspectable math (‘Fraction‘ rational cooling curve, integer Watts/kg/bps/ms, parameterized witness explanations).

Slide 2: Fail-Closed Validation & Defeating Legacy Defects

- **Strict Schema & Type Discipline (‘B2‘, ‘F3‘, ‘H06‘, ‘H13‘)**:
  - Rejects ‘NaN‘, ‘Infinity‘, negative or zero budgets, booleans used as numbers (‘True != 1‘), string booleans (‘"false"‘), unknown fields, dependency cycles, and invalid cooling coupling (‘cooling_water_w + cooling_air_mast_w == sustained_power_w‘).
  - Reproduces and defeats all **10 adversarial input flaws** (‘resource_engine_repro.json‘) and all **20 verifier mutation classes** (‘false_accepts = 0‘).
- **Six Declarative Guest Cards (‘G1‘-‘G6‘) with Field-Level Provenance**:
  - Official ‘G1‘ (‘45 kg‘, ‘1.5 kW‘ air), ‘G2‘ (‘3,500 kg‘, ‘20/6 kW‘), ‘G3‘ (‘5,500 kg‘, ‘25/1 kW‘), ‘G4‘ (‘140+850 = 990 kg‘ per errata, ‘35/10/2 kW‘), ‘G5‘ (‘3,000+25 = 3,025 kg‘, ‘45/8 kW‘), plus custom civil card ‘G6‘ (‘420 kg‘, all fields ‘ASSUMED‘).

Slide 3: Scenario A (D1) -- 8-Class Admission & Errata Proof

- **Organizer Errata Verified (‘DAT-02‘, ‘DAT-03‘)**:
  - With ‘G4 = 990 kg‘, exactly **3 of 5** four-guest combinations exceed ‘10,000 kg‘: ‘{G1,G2,G3,G4}=10,035 kg‘, ‘{G1,G2,G3,G5}=12,070 kg‘, ‘{G2,G3,G4,G5}=13,015 kg‘.
  - The 2 mass-feasible sets ‘{G1,G2,G4,G5}=7,560 kg‘ and ‘{G1,G3,G4,G5}=9,560 kg‘ prove that **passing mass does not imply passing placement**: ‘G5‘’s antenna collides with ‘G1‘ on ‘mast_3‘ (‘1.5+1.2=2.7 kW > 2.0 kW‘ air) and with ‘G4‘ drone clearance on ‘hangar_roof‘!
- **14-Day Mission Envelope (‘A.10‘) & Conditioned Availability**:
  - Rejects pier-only comfort (‘16 deg C‘); propagates ‘A.10‘ offshore thermal/sea-state restrictions directly into the final ‘YES‘ / ‘YES_IF‘ / ‘NO‘ verdict.

Slide 4: Scenario B (D3) -- Live Arbitration Under Safety Deadlines

- **The Conflict**: ‘G3‘ AUV is executing a ‘40 min‘ quiet acoustic survey (‘SURVEY_QUIET‘, ‘1 kW‘, ‘3 kn‘) when ‘G4‘ UAV reports **‘18%‘ battery (‘<20%‘ emergency -> ‘12 min / 720 s‘ max endurance)** requiring a turn into ‘22 kn‘ wind and ‘35 kW‘ recovery/charging power.
- **Non-Negotiable Safety Invariant (‘ARB-01‘..‘04‘)**:
  - Candidate plans are generated and filtered against hard physical/weather/deadline constraints before scoring by ‘priority_profile‘ weights; touchdown (‘<= T+720 s‘) is separated from the 20-min (‘1,200 s‘) ‘35 kW‘ hangar recharge.
  - **‘rescue_primary‘**: Issues ‘ACOUSTIC_HOLD‘ immediately at $T=480\text{ s}$, receives ACK at $T=495\text{ s}$, turns into wind, and lands ‘G4‘ at $T=615\text{ s}$ (‘585 s‘ safety margin before $T=1,200\text{ s}$ deadline).
  - **‘survey_primary‘**: Completes ‘240 s‘ of survey pass, issues ‘ACOUSTIC_HOLD‘ at $T=720\text{ s}$, receives ACK at $T=735\text{ s}$, turns into wind, and lands ‘G4‘ at $T=855\text{ s}$ (‘345 s‘ safety margin).

Slide 5: Scenario B (D4) -- Graceful Degradation & Thermal Binding

- **Honest Power vs. Thermal Accounting (‘DEG-01‘..‘05‘)**:
  - When generator reserve drops from ‘200 kW‘ to ‘100 kW‘, ‘G3 (1 kW survey) + G4 (35 kW) + G5 (45 kW) = 81 kW‘ actual draw ($\le 100\text{ kW}$, **zero fabricated instantaneous deficit**).
  - With G3’s firm ‘25 kW‘ turnaround recharge reservation (‘25 + 35 + 45 = 105 kW > 100 kW‘), the ‘5 kW‘ firm deficit triggers **‘G5‘ mode step-down (‘45 kW‘ $\rightarrow$ ‘8 kW‘ ‘DEGRADED_70PCT_LOSS‘)**, saving ‘37 kW‘ power and reducing committed load to ‘68 kW‘ (‘32 kW‘ free margin) while retaining ‘30%‘ capability.
  - At $30\,^\circ\text{C}$ seawater with -40% cooling loop loss (‘72 kW‘ thermal capacity vs ‘100 kW‘ power), **chilled-water cooling binds first** (‘79.8 kW > 72 kW‘) and steps down ‘G5‘ (‘79.8 kW‘ $\rightarrow$ ‘43.5 kW‘).

Slide 6: Scenario C -- Crew Regimes (14 / 6 / 0) & Resilience

- **Crew Regime Enforcement (‘CRW-01‘..‘03‘)**:
  - **14 Crew**: ‘9‘ deck/payload crew $\rightarrow$ ‘2‘ concurrent deck operations (‘G2‘ + ‘G4‘) and at-sea vehicle replacement ($A’$ window: ‘G4‘ ‘3‘ crew / ‘30 min‘; ‘G3‘ ‘4‘ crew / ‘45 min‘) permitted.
  - **6 Crew**: ‘2‘ deck crew $\rightarrow$ single serialized manual launch/recovery permitted; at-sea vehicle replacement (‘min_crew >= 3‘) blocked.
  - **0 Crew**: Zero physical installations or manual deck evolutions at sea (‘CRW-02‘); autonomous software re-tasking, mode step-down, and load shedding only.
- **Fault Resilience (‘SAF-01‘, ‘RUN-03‘, ‘RUN-04‘)**:
  - ‘STOP‘/‘HOLD‘/‘SECURE‘ measured in ‘< 1 ms‘ ($\le 200\text{ ms}$ budget); stale feeds clamp to $30\,^\circ\text{C}$ cooling; ‘+20%‘ overconsumption auto-throttled.

Slide 7: Dual-Platform Architecture & Live Ventuno Q Parity

- **Decoupled Laptop + Edge Board Execution (‘B1‘, ‘DEP-01‘..‘03‘)**:
  - **MacBook Pro (‘arm64‘, Python 3.13.15)**: Full test suite, event-driven simulator, 90 s compressed demo, and stateful 2D operator console (‘http://127.0.0.1:8765‘).
  - **Arduino Ventuno Q (‘training06‘, Qualcomm Dragonwing IQ8 ‘aarch64‘, 16 GB RAM, Ubuntu 24.04.4)**: Live SSH deployment at ‘/home/arduino/sdv-resource-governor‘.
  - **Bit-for-Bit Semantic Parity**: Identical ‘campaign_semantic_sha256‘ verified across ‘arm64‘ macOS and live ‘aarch64‘ Ventuno Q (‘training06‘).

Slide 8: Single-Screen 2D Operator Console & Fail-Closed Evidence

- **Stateful Single-Screen 2D Operator Console (‘B10‘, ‘H12‘)**:
  - Persistent ‘GovernorSession‘ (‘snapshot_id‘, ‘state_version‘, ‘POST /api/reset‘), live 8-class budget gauges, 2D vessel bay/mast/hangar/roof/rack map, 90 s compressed Scenario B timeline scrubber, instant ‘survey_primary‘ $\leftrightarrow$ ‘rescue_primary‘ switcher, perturbation injector, and live Scenario A guest card editor.
- **Fail-Closed Evidence Verifier (‘B8‘, ‘H05‘)**:
  - Recomputes all summary metrics and semantic hashes from raw JSON logs, verifies all 49 acceptance criteria ‘<testcase>‘ nodes in ‘junit.xml‘, and rejects **20 / 20** adversarial mutations (‘false_accepts = 0‘).